සියලුම හිමිකම් ඇවිරිණි/முழுப் பதிப்புரிமையுடை யது/ All Rights Reserved]

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கல்விப் பொதுத் தராதரப் பத்திர (உயர் தர)ப் பரீட்சை, 2027 ஓகஸ்ட்

General Certificate of Education (Adv. Level) Examination, August 2027

භෞතික විද්‍යාව I
பௌதிகவியல் I
Physics I

Unit 02-Kinematics
Model Paper -01

01 E I

Name: *Institute.:*

Instructions:

- *This question paper consists of **50** questions in **12** pages.*
- *Answer all the questions*
- *Write your **Institute** in the space provided in the answer sheet.*
- *Read the instructions given on the back of the answer sheet carefully.*
- *In each of the questions **1** to **50**, pick one of the alternatives from* (1), (2), (3), (4), (5) *which is correct or **most appropriate** and **mark your response on the answer sheet with a cross (X) in accordance with the instructions given on the back of the answer sheet**.*

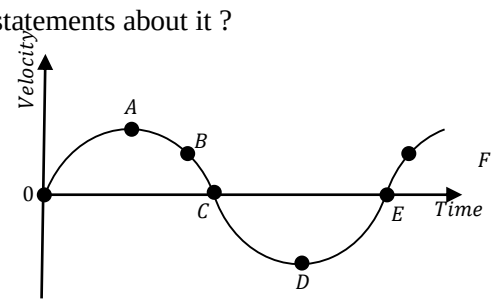
Use of calculators is not allowed

(Acceleration due to gravity, $g = 10\ \mathrm{N\ kg^{-1}}$) Duration :- **2 hrs**

1. A stone falls freely from rest and the total distance covered by it in the last second of its motion equals the distance covered by it in the first three seconds of its motion. The stone remains in the air for,
   (1) 6 s (2) 5 s (3) 7 s (4) 4 s (5) 3 s

2. A car starting from rest and travelling with uniform acceleration obtains a velocity $5\mathrm{m\ s^{-1}}$ in 2 s. After how many seconds will the car attain a velocity of $15\mathrm{m\ s^{-1}}$?
   (1) 6 s (2) 4 s (3) 8 s (4) 15 s (5) 10 s

3. A v-t graph of a moving particle is given below .what are the true statements about it ?
   (A) The particle is moving in to the same direction aat A and D
   (B) The particle is decelerating at B and accelerating at F
   (C) It changes the direction at C and E
   (1) Only (A) and (B) are correct
   (2) Only (A) and (C) are correct
   (3) Only (B) and (C) are correct
   (4) Only (C) and (D) are correct
   (5) All (A), (B) and (C) are correct

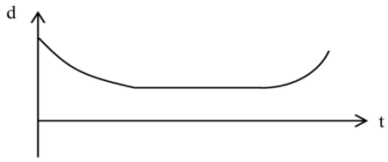


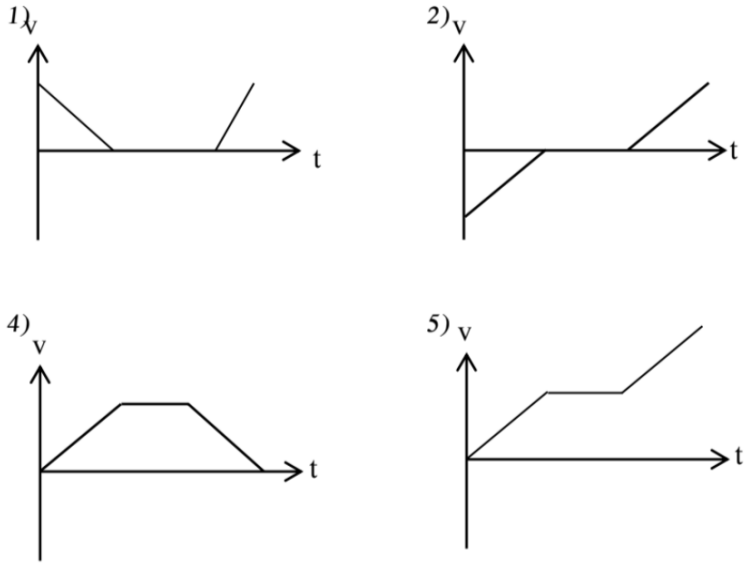
4. The following statements are about a town X
   (A) The town is a five mile drive along the winding country road.
   (B) The town sits at an altitude of 940 m.
   (C) The town X is ten miles north of the town Y.
   (1) only (A) is true. (2) only (B) is true (3) only (C) is true.
   (4) only (B) and (C) are true. (5) only (A) and (C) are true.

5. A rocket, initially at rest, is fixed vertically with an upward acceleration of $10\ \mathrm{m\ s^{-1}}$. At an amplitude of 0.5 km, the engine of the rocket cuts off. Its maximum altitude will be (Ignore the change in gravitational acceleration with altitude)
   (1) 1.9 km (2) 0.5 km (3) 1.5 km (4) 1.0 km (5) 1.6 km

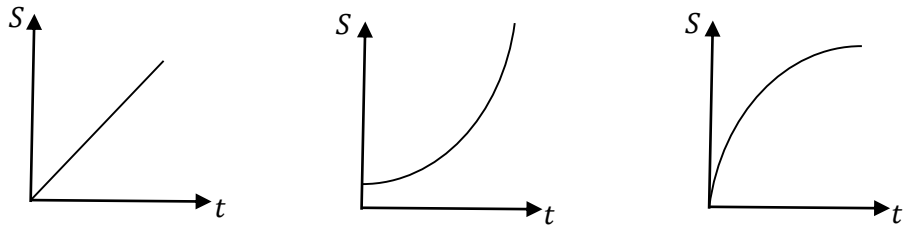
6. A ball is projected horizontally with a velocity of $5\ \mathrm{m\ s^{-1}}$ from the top of a building, 20 m high. The time the ball will take to hit the ground is
   (1) $\sqrt{2}$ s (2) 2 s (3) $\sqrt{3}$ s (4) 3 s (5) 5 s

7. A body displacement – time graph is shown in the figure. The corresponding speed (v) – time (t) graph is best represented by





8. Following are displacement (s) vs time (t) graphs. The graphs that show a positive acceleration, negative acceleration and zero acceleration respectively are,

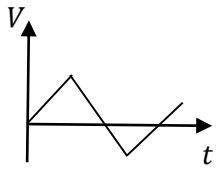


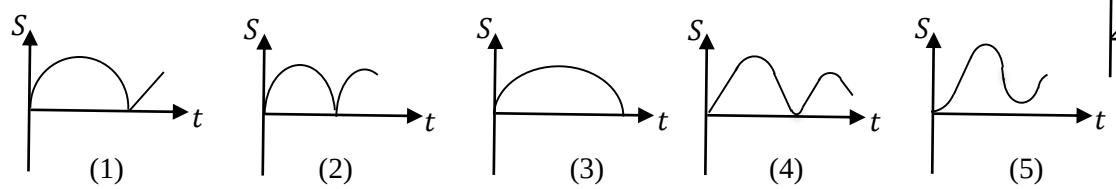
(A) (B) (C)

   (1) (A), (B), (C) (2) (B), (C), (A) (3) (C), (A), (D) (4) (C), (B), (A) (5) (A),(C),(B)

9. Considered the statement given about linear motion of an object
   (A) The displacement, velocity and acceleration always act in the same direction
   (B) When the velocity is zero, displacement is also zero
   (C) A velocity can exist where the acceleration is zero
   (D) When an object moves backward, the displacement of the object is always negative,
   Select the statements
   (1) Only (A) is correct (2) Only (B) is correct (3) Only (C) is correct
   (4) Only (C) and (D) are correct (5) All (A), (B) and (C) are correct

10. Velocity time graph of a motion is shown in the figure its displacement time graph best represented by

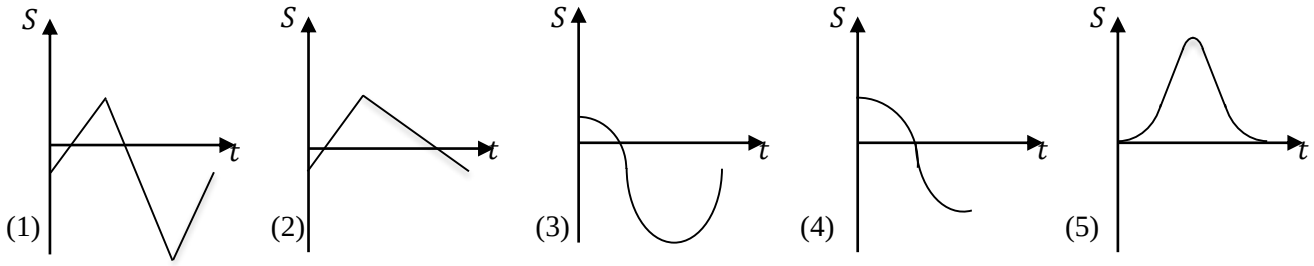




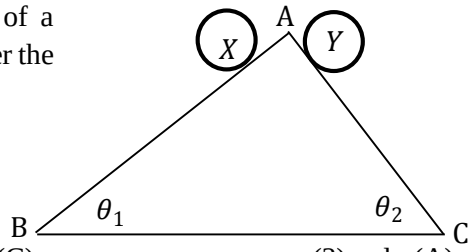
(1) (2) (3) (4) (5)

11. Select the true statement about motion of an object
(A) There can be no velocity when the acceleration is zero
(B) Reversal of direction of velocity means reversal of direction of motion
(C) In the reversal of an object, the sign of displacement in reversed
(1) Only (A) (2) Only (B) (3) Only (A) and (B)
(4) Only (B) and (C) (5) All (A), (B) and (C)

12. A car accelerates from the rest on a straight road. A short time later, the car decelerates to a stop and then returns to its original position in a similar manner, by speeding up and then slwing to a stop. Which of the following five displacement versus time graphs best describes the motion

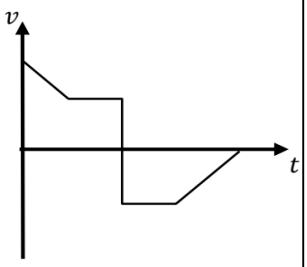


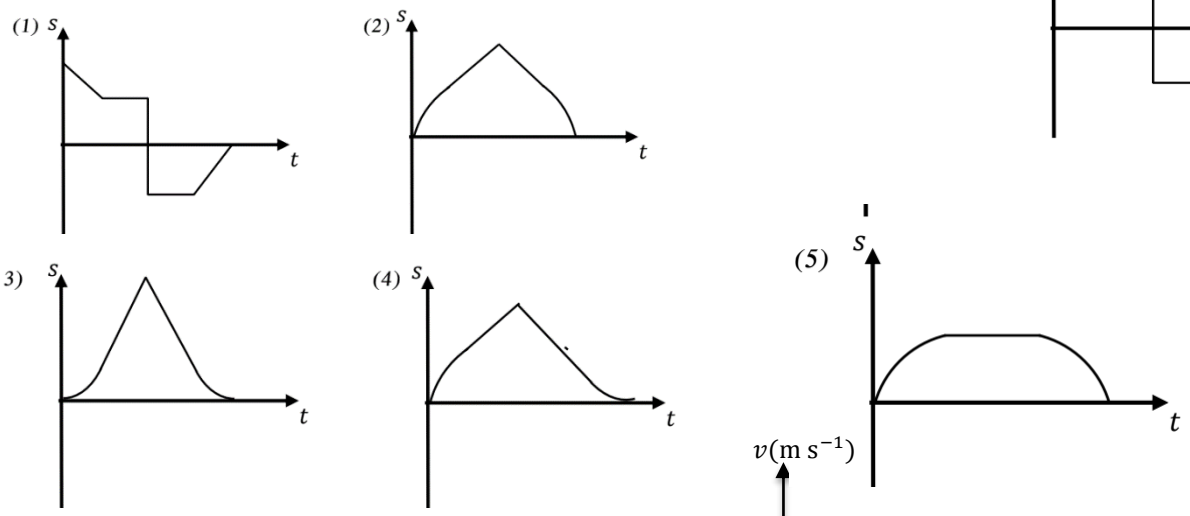
13. Two particles (X, Y) of equal density are placed at the top of a smooth inclined plane (where $\theta_2 > \theta_1$) and left freely. Consider the following statements
(A) The velocities of X and Y are equal at B and C
(B) Acceleration of Y is greater than the acceleration of X
(C) Y reaches the bottom at the first
True statement/s is/are



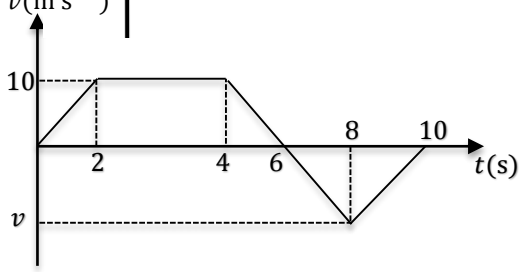
(1) only (B) (2) only (B) and (C) (3) only (A) and (B)
(4) only (A) and (C) (5) All (A), (B) and (C

14. The graph shown in the figure represents the velocity (v) – time (t) curve for an object. The corresponding displacement time curve is best represented by





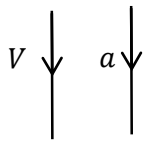
15. Velocity (v) – time (t) graph of a vehicle moving in a straight path is shown in the figure. What is the average speed of the object?
(1) $2\ \mathrm{m\ s^{-1}}$ (2) $4\ \mathrm{m\ s^{-1}}$ (3) $6\ \mathrm{m\ s^{-1}}$
(4) $8\ \mathrm{m\ s^{-1}}$ (5) $10\ \mathrm{m\ s^{-1}}$

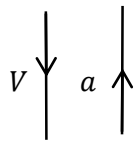


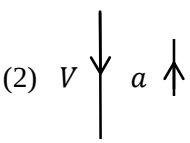
16. An object is projected horizontally, at a speed v from the top of a tower of height H. Total time of flight (T), horizontal range (R) and the speed of the projectile at the landing (V) are

	T	R	V
(1)	$\sqrt{\frac{2H}{g}}$	$v\sqrt{\frac{H}{g}}$	$\sqrt{v^2+2gH}$
(2)	$\sqrt{\frac{H}{g}}$	$v\sqrt{\frac{H}{g}}$	$\sqrt{v^2+2gH}$
(3)	$\sqrt{\frac{H}{2g}}$	$v\sqrt{\frac{2H}{g}}$	$\sqrt{v^2+gH}$
(4)	$\sqrt{\frac{2H}{g}}$	$v\sqrt{\frac{2H}{g}}$	$\sqrt{v^2-2gH}$
(5)	$\sqrt{\frac{2H}{g}}$	$v\sqrt{\frac{2H}{g}}$	$\sqrt{v^2+2gH}$

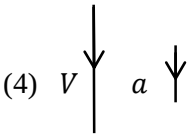
17. A tea strainer was slowly dropped by a man from a point above the ground. The given figure shows the vector representation of the instantaneous velocity V and acceleration a of the tea strainer, at a certain instant during its motion. If air resistance is considered for its motion, which of the following vector diagrams correctly depicts its instantaneous velocity V and acceleration a, after it has fallen for some more time?

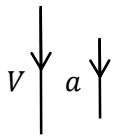


(1) 

(2) 

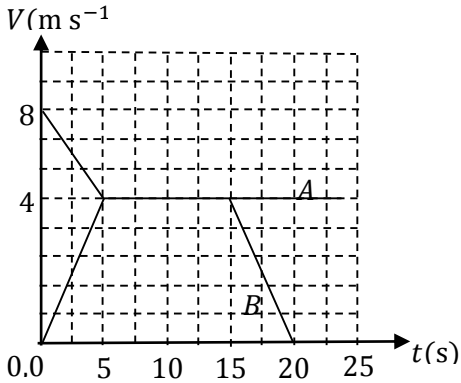
(3)

(4) 

(5) 

18. A train covers half of its journey with a speed of 20 m s^{-1} and the other half with a sped of 30 m s^{-1}. The average speed of the train during the whole journey is,
(1) 24 m s^{-1} (2) 25 m s^{-1} (3) 26 m s^{-1} (4) 27 m s^{-1} (5) Zero

19. When the signal light on a straight road turns green, vehicle A begins to move with a uniform acceleration starting from rest and then continues its motion at a uniform velocity. When vehicle A begins to move, vehicle B passes that point with a velocity of 8 m s^{-1}.Then vehicle B moves with a uniform deceleration for 5 seconds in the same direction as A, and then continues to move at a uniform velocity V of both vehicles with time t.At the moment when vehicle B comes to rest, vehicle A is



(1) 10m distance in front of B
(2) 10m distance behind B
(3) 20m distance behind B
(4) 20m distance in front of B
(5) 5m distance behind B

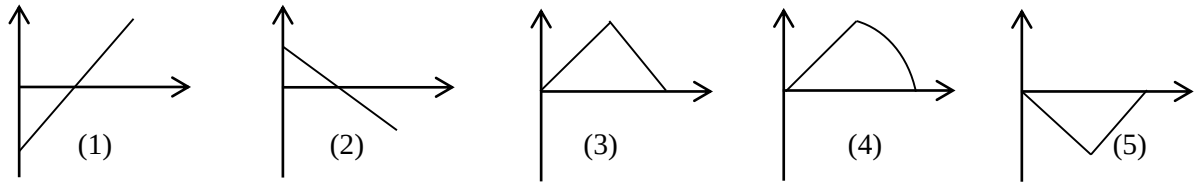
20. When two particles P and Q move towards each other along a straight line with constant speeds, they approach each other by 7 m in every second. When they move in the same direction along a straight line with their same initial speeds, they move away from each other by 3 m in every second. What are the speeds of P and Q?
(1) $4ms^{-1}$, $3ms^{-1}$ (2) $4ms^{-1}$, $2ms^{-1}$ (3) $5ms^{-1}$, $3ms^{-1}$
(4) $5ms^{-1}$, $2ms^{-1}$ (5) $3ms^{-1}$, $2ms^{-1}$

21. An object released at rest completed 19% of the total height in its last second of the motion. The total height of the motion is
(1) 405 m (2) 500 m (3) 1000 m (4) 250 m (5) 95 m

22. A body moving in a straight line with an initial velocity u and a constant acceleration a, covers a distance of 40 m in the 4th second and a distance of 60 m in the 6th second. The values of u and a respectively are
(1) $10\ \mathrm{m\ s^{-1}}, 5\ \mathrm{m\ s^{-2}}$ (2) $10\ \mathrm{m\ s^{-1}}, 10\ \mathrm{m\ s^{-2}}$ (3) $5\ \mathrm{m\ s^{-1}}, 5\ \mathrm{m\ s^{-2}}$
(4) $5\ \mathrm{m\ s^{-1}}, 10\ \mathrm{m\ s^{-2}}$ (5) $15\ \mathrm{m\ s^{-1}}, 10\ \mathrm{m\ s^{-2}}$

23. A bowler throws a ball at $100\ \mathrm{ms^{-1}}$ speed with an inclination to the horizontal plane such that the maximum range is attained. Here, if the horizontal range attained by the ball (R) and the maximum height attained by the ball is h , then the ratio $\frac{R}{h}$ is
(1) $\frac{1}{4}$ (2) 2 (3) $\frac{1}{2}$ (4) 4 (5) 5

24. A stone dropped from a certain height above a muddy pond. The stone freefalls through air , gets immersed into the mud and then decelerates uniformly to become stationary . The velocity - time graph of this motion is best represented by

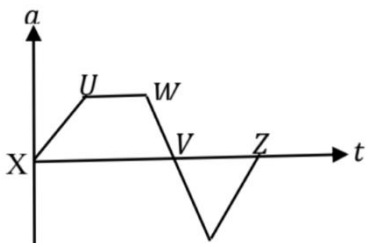


25. A ball is kicked from point P on level ground which initially travels at 45° to the horizontal. The ball reaches its maximum height after a time of 2 s. What is the maximum height reached and range of the motion? (Air resistance can be ignored.)
(1) 20 m, 80 m (2) 40 m, 40 m (3) 45 m, 180 m (4) 20 m, 80 m (5) 20 m, 80 m

26. A car starts moving along a line first with acceleration $a = 5\ \mathrm{m\ s^{-2}}$ starting from rest then uniformly and finally decelerating at the same rate $a = 5\ \mathrm{m\ s^{-2}}$ come to rest with a total time for motion of 25 s. The average velocity during the time is $72\ \mathrm{kmh^{-1}}$, how long does the particle move with uniform velocity.
(1) 5 s (2) 10 s (3) 15 s (4) 20 s (5) 12.5 s

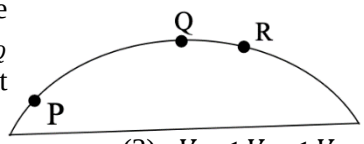
27. A stone thrown vertically upwards flies past a small window one second after it was thrown upward, and after 3 seconds in its way downward. The height of the window above the ground is,
(1) 5 m (2) 10 m (3) 15 m (4) 20 m (5) 25 m

28. A car is travelling along a straight road. The graph shows the variation with t of its acceleration (a) during a part of the journey. At which point on the graph corresponds to the greatest velocity and greatest displacement respectively?



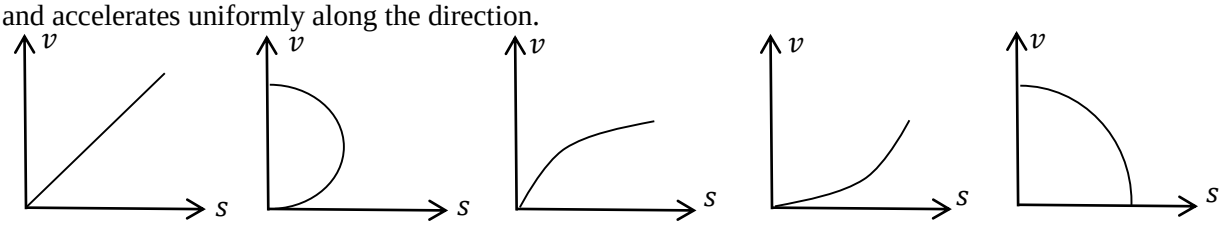
(1) V, X (2) V, Z (3) X, Y
(4) X, Z (5) W, X

29. P, Q and R, are three points on the path of a ball which is projected at an angle and movies under gravity. The air resistance on the ball is negligible. If V_P, V_Q and V_R are the speeds of the ball at the points P, Q and R respectively, the correct relationship among them is,



(1) $V_P < V_Q < V_R$ (2) $V_P < V_Q < V_R$ (3) $V_P < V_Q < V_R$
(4) $V_P < V_Q < V_R$ (5) $V_P < V_Q < V_R$

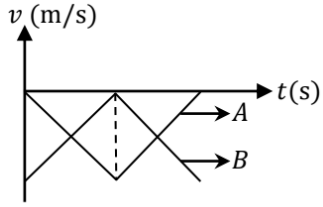
30. Which is the correct displacement (S) – velocity (v) graph for the motion of an object, which starts from rest and accelerates uniformly along the direction.

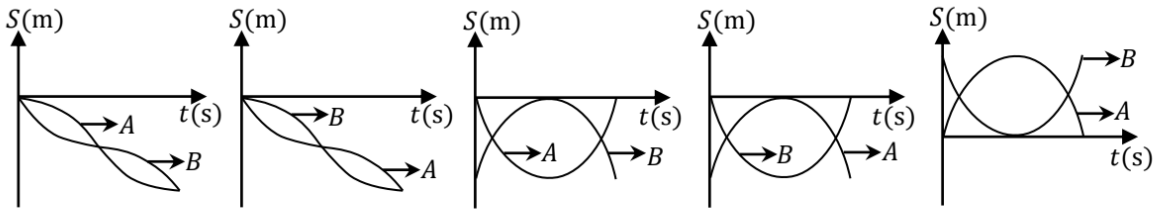


31. A ball is dropped from the top of a building of height h at time $t = 0$. If the ball reached the ground in time T, the height of the ball from the ground at time $t = \frac{T}{2}$ is,

(1) $\frac{h}{8}$ (2) $\frac{h}{4}$ (3) $\frac{h}{3}$ (4) $\frac{h}{2}$ (5) $\frac{3h}{4}$

32. The figure below shows the velocity – time graph of 2 objects A and B drawn on the same axis. Which of the following graphs best represents the displacement of two objects with time?





(1) (2) (3) (4) (5)

33. An object A is projected vertically upwards at a speed of $100\ \mathrm{m\ s^{-1}}$ from ground level. Simultaneously, another object B dropped freely from a location linearly upwards from the projected one. After 3 seconds, the distance between A and B is 75m. What is the distance between the ground and the point at where B was released,

(1) 325 m (2) 375 m (3) 385 m (4) 400 m (5) 410 m

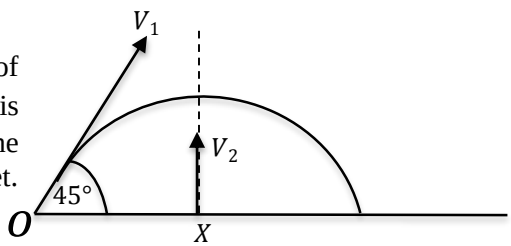
34. Water in a horizontal straight river a flowing with constant velocity v. A person who could swim at u velocity in still water, swims along the flowing direction of the river with same efficiency, He starts swimming from a specific point in the river and swims up to a certain distance. Then he immediately turns around and swims back to the initial point with same efficiency, What is the average speed of his complete motion?

(1) u (2) $\sqrt{u^2 - v^2}$ (3) $\frac{(u-v)^2}{u}$ (4) $\frac{(u+v)^2}{u}$ (5) $\frac{u^2-v^2}{u}$

35. A train of 150 m length is going towards north direction at a speed of $10\ \mathrm{m\ s^{-1}}$.A parrot flies at a speed of $5\ \mathrm{m\ s^{-1}}$towards south direction parallel to the railway track. The time taken by the parrot to cross the train is equal to

(1) 12 s (2) 8 s (3) 15 s (4) 30 s (5) 10 s

36. A particle is projected from point O with a velocity v_1 at an angle of 45^0 with the horizontal. At the same instant another particle is projected verticaly upwards from point X with a velocity v_2. At the maximum height of the first particle, both the particles barely meet. The ratio $\frac{v_2}{v_1}$,

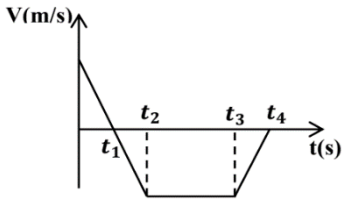


(1) $\sqrt{2}$ (2) $\frac{1}{\sqrt{2}}$ (3) $\frac{1}{2}$ (4) $2\sqrt{2}$ (5) 2

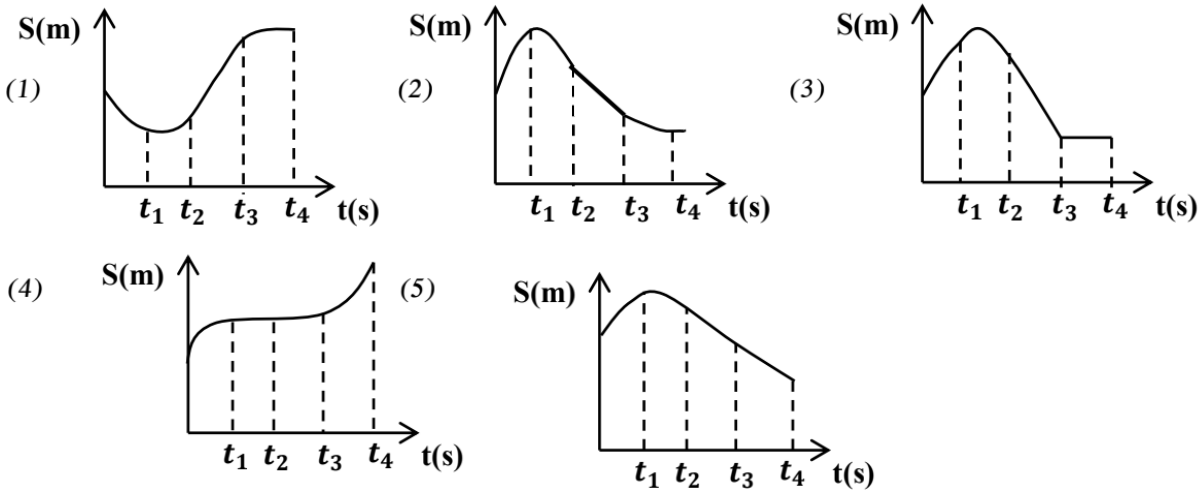
37. An object of mass m is projrcted vertically upwards and another object of $2m$ is projected upwards with an angle θ to the horizontal.If both objects reached their maximum height at the same time period. The ratio between maximum heights of both objects is

(1) $1:1$ (2) $2:1$ (3)$1:2$ (4) $\cos\theta : 1$ (5) $1:\cos\theta$

38. Three particles A, B and C were projected with same speed at angles $15°, 45°$ and $75°$ respectively from a point. Their ranges are S_A, S_B and S_C respectively and their maximum vertical heights are H_A, H_B and H_C respectively. Find the correct relationship.
(1) $S_A = S_C < S_B,\ H_A > H_B > H_C$
(2) $S_A = S_C > S_B,\ H_A = H_C > H_B$
(3) $S_A > S_B > S_c,\ H_A > H_B > H_C$
(4) $S_A > S_B > S_C,\ H_A = H_C > H_B$
(5) $S_A = S_C < S_B,\ H_A < H_B < H_C$



39. The corresponding displacement time graph for the velocity time graph is best represented by



40. Graphs show the variation of horizontal displacement (S_x) and vertical displacement (S_y) of an object thrown at an angle θ with horizontal with time θ equals?
(1) $90°$ (2) $30°$ (3) $60°$ (4) $45°$ (5) 75

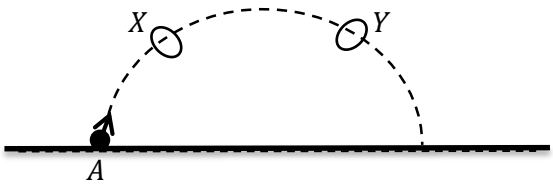
41. A rescue plane flies at $198\ \text{kmh}^{-1}$ and constant height $h = 500$ m towards a point directly over a victim, where a rescue capsule is to land. The angle made by the pilot's line of sight to the victim with the downward vertical when the capsule is released is
(1) $\tan^{-1}\frac{550.5}{500}$ (2) $\tan^{-1}\frac{555.5}{500}$ (3) $\tan^{-1}\frac{500}{555.5}$ (4) $\tan^{-1}\frac{500}{550}$ (5) $\tan^{-1}\frac{550}{500}$

42. A boy stands 4 m away from a vertical wall and throws a ball. The ball leaves the boy's hand at $h = 1.05$ m above the ground with initial velocity $u = 10\sqrt{2}\ \text{m s}^{-1}$ at an angle of $45°$ with the horizontal. After striking the wall elastically, the ball rebounds. The distance from the foot of the wall to the point where the ball hits the ground is
(1) 15 m (2) 16 m (3) 17 m (4) 20 m (5) 25 m

43. Figure shows three kicks in the air, all reach same height. Air resistance could be neglected. Consider the following statements.
A- Time of flight for all three kicks are the same.
B- vertical components of the initical velocities are same for all 3 kicks.
C- Horizontal component of initial velocity is the greatest for R
Which of the following statements is/are correct?
(1) only B (2) only C (3) only A and B
(4) only B and C (5) All A, B and C

44. An object is projected vertically upwards at initial velocity u.It takes 6 s to reach maximum height. The ratio between the distances moved by the object in 1st second and 7^{th}second of its motion is
(1) 1:11 (2) 1:1 (3)11:1 (4) 1:2 (5) 2:1

45. A particle flung from point A in the horizontal plane passes through two rings located at the same horizontal level. It took one second for the particle to go from A to X and X to Y each. What heights were the rings positioned from the ground level?



(1) 30 m (2) 20 m (3) 15 m (4) 10 m (5) 5 m

46. As shown in the diagram, A is projected with a velocity V_1 from P. At the same time another object B is projected up with a velocity V_2 from Q. They will meet each other at R. The ratio of $\frac{V_1}{V_2}$ is

(1) 1 (2) 2 (4) $\frac{1}{2}$ (4) $\frac{1}{4}$ (5) 4

47. Consider the following statements about graphs of motion.

(A) The gradient of a tangent drawn at a particular point on a displacement – time graph gives the velocity of the moving object at that instance but area under the graph provides no details about the motion.

(B) The gradient of a tangent drawn at a particular point on a velocity – time graph gives the acceleration of the moving object at that instance.

(C) The gradient of a tangent drawn at a perpendicular point on an acceleration – time graph gives no detail about the motion but the area between the curve and the time axis provides change in velocity of the object.

(D) The gradient of a tangent drawn at a particular point on an acceleration-time graph gives no details about the motion but the area between the curve and the time axis provides change in velocity of the object.

Of the above statements

(1) only (A), (C) and (D) are true
(2) only (B), (C) and (D) are true
(3) only (A), (B) and (C) are true
(4) only (A), (B) and (D) are true
(5) all (A), (B), (C) and (D) are true

48. An object slides down along an inclined plane of inclination θ to the horizontal and falls freely from its lower end and then it's moving under gravity. The value for θ for its horizontal and vertical components of velocity to be equal at a position during the free fall is

(1) 30° (2) 45° (3) 50° (4) 60° (5) 75°

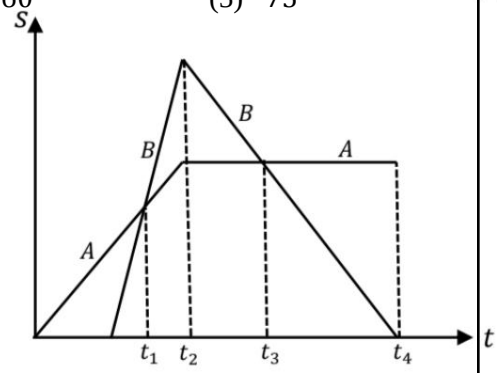
49. Diagram shows the displacement (s) – time (t) graph drawn for the displacement measured from the moment when two vehicles A, B moving in a straight path pass a particular point P. Consider the following statements,

(A) – Acceleration of B is higher than that of A

(B) – At time $t = t_3$, A and B meet together for the second time.

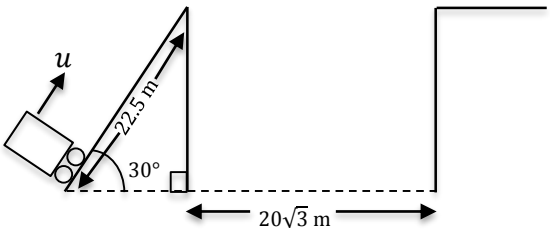
(C) – After the time $t = t_3$, velocities of A and B are opposite to one another

Which of the above statements is/are true?



(1) (A) only (2) (B) only (3) (A) and (C) only
(4) (B), (C) only (5) (A), (B) and (C) all

50. A trolley is thrown with velocity u at the base of a smooth inclined plane of length 22.5 m and 30° inclination with the horizontal. Trolley reaches the tip of the inclined plane and undergoes an inclined projectile motion to land at the horizontal level at a distance of $20\sqrt{3}$m from the inclined plane. What is the minimum value of u needed to reach this horizontal level?



(1) 10 m s^{-1} (2) 15 m s^{-1} (3) 20 m s^{-1} (4) 50 m s^{-1} (5) 25 m s^{-1}